

Homework 1: Surface Pressure, Lift, and Pressure Drag

NACA 2412 at $\alpha = 5$ degrees

Name: _____

UMass ID: _____

Due: _____

Section: _____

Purpose

In this homework you will compute the two-dimensional aerodynamic force produced by a pressure distribution on a NACA 2412 airfoil. In Question 1, the geometry and pressure are continuous functions and you will evaluate surface integrals. In Question 2, the same flight condition is represented by tabulated data and you will replace the integrals by finite differences and panel sums.

Model statement. The supplied pressure distribution is a calibrated instructional surrogate for attached flow over a NACA 2412 section at $\alpha = 5$ degrees. It reproduces a physically reasonable suction peak and a lift coefficient near the historical NACA 2412 trend. It is *not* pressure-tap measurement data or a universal pressure law.

Given conditions and conventions

Quantity	Value	Quantity	Value
Airfoil	NACA 2412	Angle of attack	$\alpha = 5.0$ deg
Chord	$c = 0.500$ m	Reference span	$b = 1.000$ m
Freestream speed	$V(\infty) = 30.0$ m/s	Density	$\rho(\infty) = 1.225$ kg/m ³
Static pressure	$p(\infty) = 101325$ Pa	Dynamic pressure	$q(\infty) = (1/2) \rho V^2$

Let $\bar{x} = x/c$ and $\bar{y} = y/c$. The x-axis points from the leading edge toward the trailing edge, and positive y points upward. Pressure coefficient is

$$C_p = [p - p_{\infty}] / q_{\infty}$$

First resolve the pressure resultant into chord-normal coefficient C_N and chord-axial coefficient C_A . Positive C_A points from leading edge to trailing edge. Rotate these coefficients into wind axes:

$$C_L = C_N \cos(\alpha) - C_A \sin(\alpha)$$

$$C_{D,p} = C_N \sin(\alpha) + C_A \cos(\alpha)$$

Only pressure forces are included. Therefore $C_{D,p}$ is **pressure drag**, not total viscous drag. Skin-friction drag cannot be recovered from the supplied pressure data.

NACA 2412 geometry used in both questions

For NACA 2412: $m = 0.02$, $p = 0.40$, and $t = 0.12$. The nondimensional mean camber line is

$$\bar{y}_c(\bar{x}) = (m/p^2)(2p\bar{x} - \bar{x}^2), 0 \leq \bar{x} \leq p$$

$$\bar{y}_c(\bar{x}) = [m/(1-p)^2][(1-2p) + 2p\bar{x} - \bar{x}^2], p < \bar{x} \leq 1$$

The closed-trailing-edge thickness distribution is

$$\bar{y}_t = 5t[0.2969 \sqrt{\bar{x}} - 0.1260 \bar{x} - 0.3516 \bar{x}^2 + 0.2843 \bar{x}^3 - 0.1036 \bar{x}^4]$$

Use the common-x approximation

$$\bar{y}_u = \bar{y}_c + \bar{y}_t, \bar{y}_l = \bar{y}_c - \bar{y}_t$$

The slopes needed in the force integral are

$$d\bar{y}_u/d\bar{x} = d\bar{y}_c/d\bar{x} + d\bar{y}_t/d\bar{x}, d\bar{y}_l/d\bar{x} = d\bar{y}_c/d\bar{x} - d\bar{y}_t/d\bar{x}$$

$$d\bar{y}_c/d\bar{x} = (2m/p^2)(p-\bar{x}), 0 \leq \bar{x} \leq p$$

$$d\bar{y}_c/d\bar{x} = [2m/(1-p)^2](p-\bar{x}), p < \bar{x} \leq 1$$

$$d\bar{y}_t/d\bar{x} = 5t[0.2969/(2\sqrt{\bar{x}}) - 0.1260 - 0.7032\bar{x} + 0.8529\bar{x}^2 - 0.4144\bar{x}^3]$$

Question 1 - Continuous surface integration [45 points]

Define the following smooth loading model on $0 \leq \bar{x} \leq 1$:

$$C_0(\bar{x}) = (1-\bar{x})\{\exp[-\bar{x}/0.010] + 0.20[1-\exp(-\bar{x}/0.020)]\}$$

$$\Delta C_p(\bar{x}) = 0.721896 [1-\exp(-\bar{x}/0.010)](1-\bar{x}) / \sqrt{\bar{x}+0.020}$$

$$C_{p,u} = C_0 - 0.80 \Delta C_p, C_{p,l} = C_0 + 0.20 \Delta C_p$$

The corresponding dimensional pressures are

$$p_u = p_{\infty} + q_{\infty} C_{p,u}, p_l = p_{\infty} + q_{\infty} C_{p,l}$$

Complete the following tasks

1a. Decode the four digits of NACA 2412. Calculate q_{∞} and state its units.

1b. Plot $\bar{y}_u$ and $\bar{y}_l$ on equal axes. On a separate graph, plot $C_{p,u}$ and $C_{p,l}$ against $\bar{x}$ using the aerodynamic convention in which the C_p axis is inverted. Identify the leading-edge suction region and the pressure difference that produces positive lift.

1c. Starting from the pressure force on each surface, show that the common-x model gives

$$C_N = \int_0^1 (C_{p,l} - C_{p,u}) d\bar{x}$$

$$C_A = \int_0^1 [C_{p,u}(d\bar{y}_u/d\bar{x}) - C_{p,l}(d\bar{y}_l/d\bar{x})] d\bar{x}$$

State the sign convention used for C_A .

1d. Evaluate C_N and C_A numerically. The thickness slope contains $1/\sqrt{\bar{x}}$ at the leading edge, but the integral is finite. Use adaptive quadrature, begin at a documented small positive value, or use $\bar{x} = s^2$. State your method and demonstrate numerical convergence using two tolerances or two discretizations.

1e. Calculate C_L , $C_{D,p}$, and the forces per unit span

$$L' = q_{\infty} C_L, D'_p = q_{\infty} C_{D,p}$$

Report coefficients to four significant figures and forces to an appropriate precision.

1f. Give two physical checks on your answer. At minimum, discuss the signs and relative magnitudes of lift and pressure drag and explain why this calculation cannot produce total viscous drag.

Question 2 - Tabulated data and discrete panel sums [45 points]

Download the companion workbook **MIE446_HW1_NACA2412_Data.xlsx**. The Data sheet contains 101 stations from $x_{bar} = 0$ to 1 and the five supplied quantities x_{bar} , y_{bar}_u , y_{bar}_l , p_u , and p_l . Do not replace the supplied values.

2a. In new columns, calculate $C_{p,u}$ and $C_{p,l}$ from the tabulated dimensional pressures.

2b. Calculate dy_{bar}_u/dx_{bar} and dy_{bar}_l/dx_{bar} at every station. For interior points use the centered difference

$$(d y_{bar}/d x_{bar})_i \text{ approximately equals } (y_{bar}_{i+1} - y_{bar}_{i-1}) / (x_{bar}_{i+1} - x_{bar}_{i-1})$$

Use one-sided differences at the first and last stations. Retain the leading-edge values; do not hide them because they are large.

2c. Form the pointwise integrands

$$I_{N,i} = C_{p,l,i} - C_{p,u,i}$$

$$I_{A,i} = C_{p,u,i}(dy_{bar}_u/dx_{bar})_i - C_{p,l,i}(dy_{bar}_l/dx_{bar})_i$$

Use the trapezoidal rule over the 100 intervals:

$$C_N \text{ approximately equals } \sum_{i=1}^{100} [(I_{N,i} + I_{N,i+1})/2] \Delta x_{bar}_i$$

$$C_A \text{ approximately equals } \sum_{i=1}^{100} [(I_{A,i} + I_{A,i+1})/2] \Delta x_{bar}_i$$

Keep each interval contribution visible in the workbook.

2d. Independently check the axial result with the direct panel line integral

$$\Delta C_{A,i} = \text{mean}(C_{p,u,i})(y_{bar}_{u,i+1} - y_{bar}_{u,i}) - \text{mean}(C_{p,l,i})(y_{bar}_{l,i+1} - y_{bar}_{l,i})$$

Use the mean of adjacent pressure coefficients on each panel and sum all 100 panel contributions. Explain why this form avoids dividing by a small Δx_{bar} .

2e. Rotate the discrete C_N and C_A into C_L and $C_{D,p}$, then calculate L' and D'_p . Compare Question 2 with Question 1 using percent difference = $100 |(discrete - continuous)/continuous|$

Comment separately on the lift difference and pressure-drag difference. Pressure drag is obtained by cancellation of two larger terms and is therefore more sensitive to discretization.

2f. Create two clear Excel charts: airfoil geometry and inverted- C_p distribution. Label axes and identify upper and lower surfaces.

Submission and engineering communication [10 points]

Submit the following through Canvas:

1. One PDF containing derivations, figures, results, units, and your comparison.
2. The completed Excel workbook with formulas visible and traceable.
3. One summary slide containing the method, final values, and one independent check.
4. A brief AI-use disclosure.

AI statement. AI tools may be used for syntax help, debugging, plotting, and checking intermediate work. You remain responsible for the derivation, spreadsheet formulas, physical interpretation, and verification. Briefly disclose what tool was used, for what purpose, and what independent check you performed. 'The AI produced this answer' is not a verification method.

Grading guide

Criterion	Points
Question 1: geometry, pressure interpretation, and force derivation	18
Question 1: integration, convergence, coefficients, forces, and checks	27
Question 2: spreadsheet organization, C_p , slopes, and integrands	18
Question 2: trapezoidal sum, direct-panel check, rotation, and comparison	27
Figures, units, summary slide, and AI disclosure	10
Total	100

References

1. E. N. Jacobs, K. E. Ward, and R. M. Pinkerton, *The Characteristics of 78 Related Airfoil Sections from Tests in the Variable-Density Wind Tunnel*, NACA Report 460. <https://ntrs.nasa.gov/citations/19930091108>
2. NASA OpenVSP Ground School, 'NACA 4-Series Airfoil.' <https://www.nasa.gov/reference/openvsp-cross-sections/>
3. NASA Glenn Research Center, 'Lift Equation.' <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/lift-equation/>